Lecture 10: The Origins of Business Cycles

Christian Wolf

MIT

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Overview

- We'll now turn to our third question: what are the **origins of business cycles**?

- Proceed in two steps:

  a) Review purely **semi-structural** literature on shock identification
  Can we identify any shocks that are credible main sources of cyclical fluctuations?

  b) How can time-series moments inform **structural business-cycle modeling**?

     (i) Business-cycle anatomy Angeletos et al. (2020)

     (ii) RANK/HANK model estimation using likelihood-based methods Justiniano et al. (2010), Auclert et al. (2020)

Outline

1. Semi-Structural Methods
   - TFP Shocks
   - Investment Technology Shocks

2. Structural Model Estimation
   - Basics of Likelihood-Based Estimation
   - Business-Cycle Anatomy
   - RANK & HANK Model Estimation

3. Summary

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Background

- For better or worse, semi-structural analysis of business-cycle origins has been largely focused on "**technology shocks**"
- Background is the history of structural macro modeling, notably Kydland-Prescott (1982)
  - Classical RBC business-cycle analysis is built around the aggregate production function

$$y_t = a_t f(k_t, \ell_t)$$

  - Core finding: in RBC models, shocks to the exogenous process driving a_t—*and only those shocks*—can generate what looks like typical aggregate business cycles
    - → Why only those shocks? immediate from flexible labor supply relation
      Recall: FOC is $-\gamma \hat{c}_t + \hat{a}_t + \alpha \hat{k}_{t-1} - \alpha \hat{\ell}_t = \frac{1}{\varphi} \hat{\ell}_t$, so co-movement requires $\hat{a}_t \neq 0$.
    - → Quantitative "success" rests on a very high Frisch elasticity that is simply implied by the chosen preference specification. We can chat more if you're interested.
- This created an entire **empirical research agenda**: Can we identify "technology shocks"? If so, do they drive cyclical fluctuations, in our FVD/FVR sense?

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Identification challenge

Q: How should we go about identifying "technology shocks"?

- **Early attempts**
  - The early literature simply used Solow residuals as a direct measure of technology shocks, e.g. Prescott (1986)
  - This of course has a host of problems: assumes specific production function and competition, doesn't allow for variation in capacity utilization, ...
- Literature thus moved ahead quickly. Will review **two more refined approaches**
  1. Galí (1999): VAR in labor productivity and employment, assume only tech. shocks can have long-run effects on productivity. What are the problems with this set-up?
  2. Basu et al. (2006): manually adjust Solow residual for time-varying utilization rates

Note: similar results are also reported in the recent max-share contribution of Francis et al. (2014).

Results

Galí (1999)

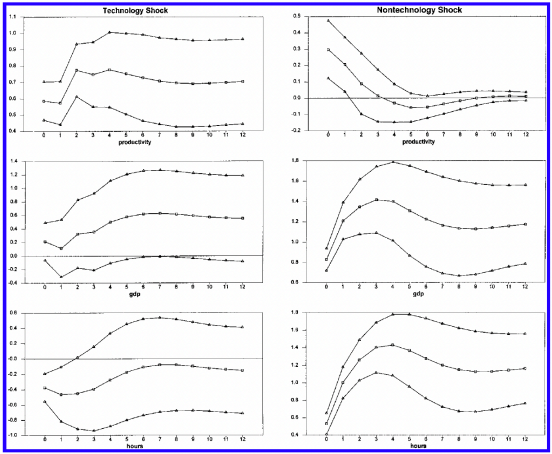
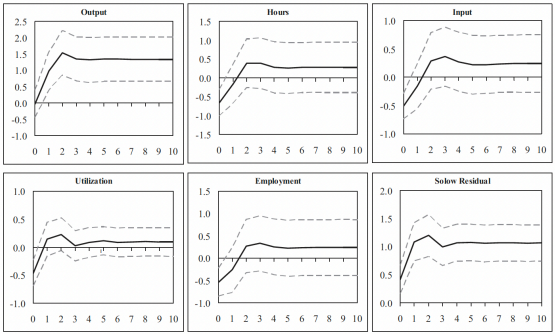


FIGURE 2. ESTIMATED IMPULSE RESPONSES FROM A BIVARIATE MODEL: U.S. DATA, FIRST-DIFFERENCED HOURS (POINT ESTIMATES AND ±2 STANDARD ERROR CONFIDENCE INTERVALS)

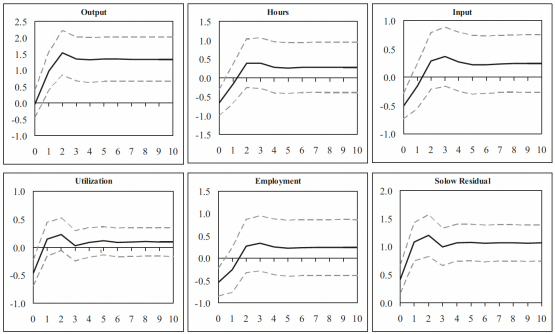
Results

Basu et al. (2006)



Results

Basu et al. (2006)



Interpretation: consistent with NK, not RBC. Implies little role for technology shocks as a source of cyclical fluctuations (since IRFs don't look like business cycles).

Technology news

- If not contemporaneous technology shocks, then maybe **news shocks**?
  Basic idea goes back to Beveridge (1909) and Pigou (1927).
  - RBC model challenge: positive news about the future make households wealthier, leading to a decline in labor supply, leading to a recession today
  - But can make it work with a couple of twists, chiefly requiring a weak wealth effect in labor supply See Jaimovich and Rebelo (2009) for the details.
- What can we see in the **data**?
  1. Beaudry-Portier (2006): bivariate VAR in stock prices and TFP, find that the same shock drives both (i) short-run stock price changes $\perp$ TFP and (ii) long-run TFP
     Important concern is non-invertibility. See Forni et al. (2014) for a discussion.
  2. Barsky et al. (2014): maximize TFP contribution at medium horizons and include larger information set $\rightarrow$ alleviates non-invertibility concerns with Beaudry-Portier

Results

Beaudry-Portier (2006)

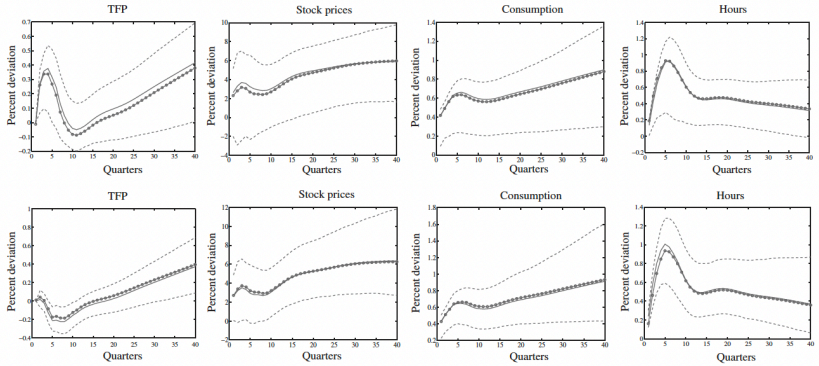


FIGURE 9. IMPULSE RESPONSES TO ε_2 AND $\tilde{\varepsilon}_1$ IN THE (TFP, SP, C, H) VECM, WITHOUT (UPPER PANELS) OR WITH (LOWER PANELS) ADJUSTING TFP FOR CAPACITY UTILIZATION

Results

Barsky et al. (2014)

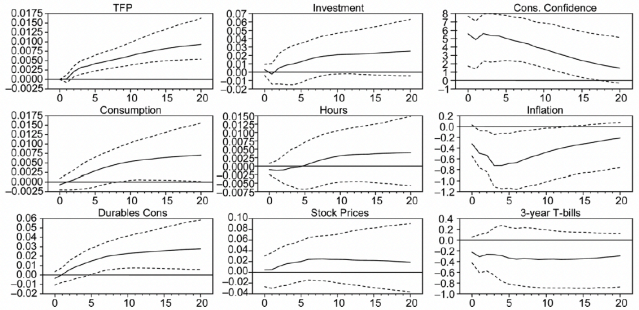


Fig. 3. Impulse responses to orthogonalized news shocks (9-variable VAR, hybrid specification).

Results

Barsky et al. (2014)

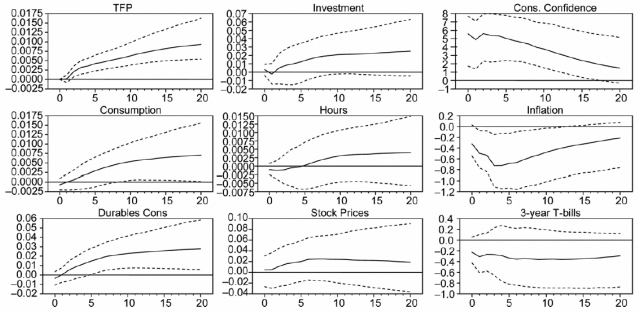


Fig. 3. Impulse responses to orthogonalized news shocks (9-variable VAR, hybrid specification).

Interpretation: improved specification is not consistent with TFP news as a main driver

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Investment technology shocks

- Alternative: maybe shocks specific to **investment technology** (rather than neutral technology shocks) are important

  Goes back to discussions in Keynes (1936).

- Will briefly review two contributions studying the importance of **IST shocks**

  1. Fisher (2006): IST analogue of Galí (1999), identifies IST shocks as the only shocks that move the relative price of investment in the long run

     Neutral technology shocks are identified via long-run labor productivity, as in Galí (1999).

  2. Ben Zeev & Khan (2015): use medium-run restrictions to identify IST news shocks, similar to Barsky et al. (2014)

Results

Fisher (2006)

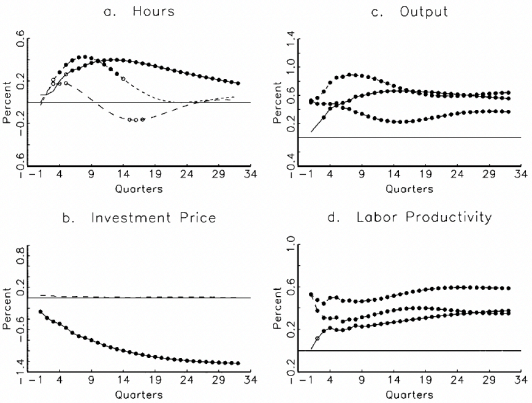
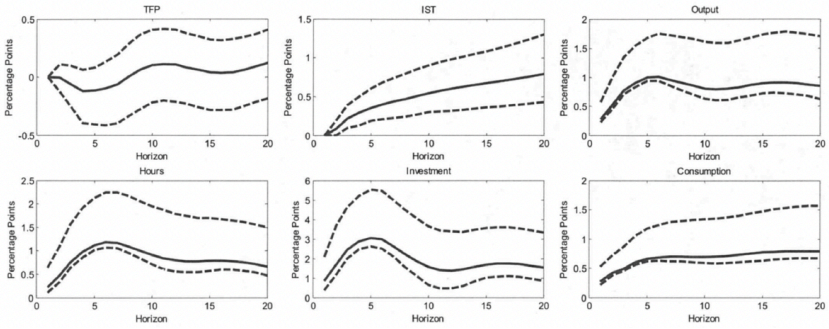


FIG. 5.—Responses to technology shocks, 1982:III–2000:IV. Solid lines denote responses to I-shocks in the five-variable system, long-dashed lines responses to N-shocks in the five-variable system, and short-dashed lines responses to N-shocks in the four-variable system. Solid circles indicate that the response is significant at the 5 percent level and open circles at the 10 percent level.

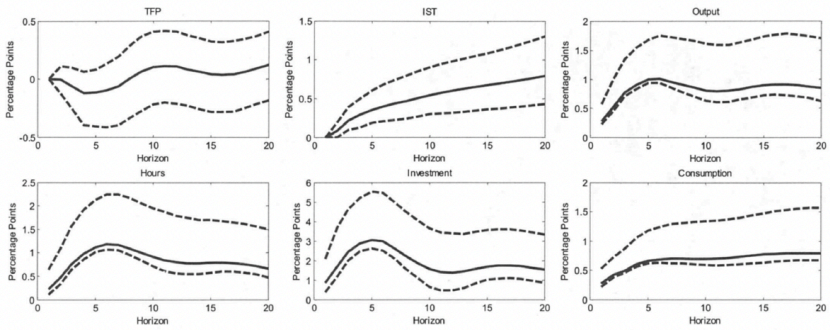
Results

Ben Zeev & Khan (2015)



Results

Ben Zeev & Khan (2015)



Interpretation: IST news shocks can account for typical business-cycle comovements

Summary

So what have we learned?

- **Main conclusions**
  - Find little support for the classical TFP shocks stressed by the old RBC literature as a source of cyclical fluctuations
  - Somewhat more evidence in favor of investment-specific technology shocks, but overall quite far from being conclusive
    E.g.: is it plausible that all medium-run IST price movements are structural IST tech. shocks?
  - More robust conclusion: important primitive business-cycle shocks must similarly be moving IST prices in the medium-run
- Recent published work on business-cycle origins leverages additional **model structure**

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Overview

- The alternative approach is **likelihood-based estimation** of a structural model
  - Implementation
    - → Write down structural model (= internal propagation mechanism) subject to several exogenous shocks (in particular incl. TFP and IST)
    - → Then find parameterization that matches U.S. time series as well as possible
  - Interpretation
    - → Origins of business cycles = shocks that, in the model, induce co-movement among aggregate time series that look like the U.S. business cycle
    - → What do you think of this approach to identification? Will this teach us much?
- Next few slides:
  1. Brief review of **likelihood-based model estimation**
  2. A diagnostic device: "**business-cycle anatomy**"
  3. Frontier estimated models: **RANK & HANK**

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Likelihood-based estimation

- Both Bayesian and classical ML estimation of structural business-cycle models rely on **likelihood evaluation** [Will do recursive here. Could also do sequence-space, using VMA(∞).]
- From **state-space model** to **likelihood**
  - A model is a parameter vector ψ giving rise to a state-space system for observables y_t:

$$\begin{aligned} y_t &= \Psi(s_t;\psi) + u_t, \quad u_t \sim F_u(\bullet;\psi) \\ s_t &= \Phi(s_{t-1},\varepsilon_t;\psi), \quad \varepsilon_t \sim F_\varepsilon(\bullet;\psi) \end{aligned}$$

    Note that for now this is slightly more general than before, allowing for non-linearities.

  - We observe data $y_{1:T}$. **Q**: What is the likelihood of $y_{1:T}$ given a parameter vector ψ? This is the key input to standard ML or Bayesian model estimation.
  - Standard approach to likelihood evaluation: filtering

A generic filter

- We can always factorize the likelihood as follows:

$$p(y_{1:T} \mid \psi) = \prod_{t=1}^{T} p(y_t \mid y_{1:t-1}, \psi)$$

- This object can in general be evaluated by proceeding as follows:
  0. Let $p(s_0) = p(s_0 \mid y_{1:0})$ be an initial distribution for the states, e.g., the stat. distribution
  1. Forecasting t given $t-1$
     a) Transition equation
     $$p(s_t \mid y_{1:t-1}) = \int p(s_t \mid s_{t-1}, y_{1:t-1}) p(s_{t-1} \mid y_{1:t-1}) ds_{t-1}$$
     b) Measurement equation
     $$p(y_t \mid y_{1:t-1}) = \int p(y_t \mid s_t, y_{1:t-1}) p(s_t \mid y_{1:t-1}) ds_t$$
  2. Updating with Bayes' theorem
     $$p(s_t \mid y_{1:t}) = p(s_t \mid y_t, y_{1:t-1}) = \frac{p(y_t \mid s_t, y_{1:t-1}) p(s_t \mid y_{1:t-1})}{p(y_t \mid y_{1:t-1})}$$

Linear model estimation & beyond

- **Linear models**
  - In linear state-space models with normal errors, easily get likelihood via Kalman filter
  - Specifically, the Kalman filter turns $p(s_{t-1} \mid y_{1:t-1})$ into $p(s_t \mid y_{1:t-1})$, $p(y_t \mid y_{1:t-1})$ and $p(s_t \mid y_{1:t})$. Stringing together T steps, we can evaluate the likelihood.
  - Model estimation then proceeds by finding ψ to maximize $p(y_{1:T} \mid \psi)$ (plus perhaps prior).
    See Fernández-Villaverde, Rubio-Ramírez, & Schorfheide (2016) for details.
- **Non-linear models**
  - Estimation of non-linear structural macro models is beyond this course
  - If you're interested: a general-purpose (but demanding) method to solve the generic filtering problem is the so-called particle filter
- **Concern**: likelihood-based estimation requires entire model to be correctly specified
  Including in particular the shock processes. Remember the "dubiously structural" shocks.

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Business-cycle anatomy

- Before presenting results from estimated business-cycle models, I will first introduce a useful (reduced-form) diagnostic device: "**business-cycle anatomy**"
  - Developed in Angeletos et al. (2020)
  - Purpose: will yield an interesting perspective on a) the origins of cycles in general & b) the limitations of the model estimation results that we'll review later
- What we'll do here:
  1. Sketch the **econometric procedure**
  2. Present the **main results**
  3. Draw some **general lessons** for theories of business cycles

Econometric procedure

- Basic idea: try to find "main shocks" that drive the various aggregate macro var's
- This is implemented using the **max-share** approach to shock identification
  - Let y_t denote a list of macro aggregates (output, consumption, unemployment, investment, inflation, interest rates, TFP ...)
  - Estimate a VAR in y_t, and invert it to get

$$y_t = C(L)Q\varepsilon_t$$

    where ε_t denotes orthogonalized unit variance shocks (may not be structural!) and Q is a rotation matrix
  - Then, for each variable y_k, find the "shock" that accounts for the largest possible share of volatility over frequencies $[\underline{\omega}, \bar{\omega}]$. That is, choose q (where $||q|| = 1$) to maximize

$$\int_{\omega\in[\underline{\omega},\bar{\omega}]} \left(\overline{C_k(e^{-i\omega})q}\; C_k(e^{-i\omega})q\right) d\omega = q' \int_{\omega\in[\underline{\omega},\bar{\omega}]} \left(\overline{C_k(e^{-i\omega})}'\; C_k(e^{-i\omega})\right) d\omega q$$

    Note: no pretense that these are "structural" shocks, but implied patterns may still be interesting

Results


Unemployment
Output
Hours Worked
Investment
Consumption
TFP
Labor Productivity
Labor Share
Inflation Rate
Nom. Int. Rate
u shock; Y shock; I shock; h shock; C shock; Shaded area: 68% HPDI.

Results

	u	Y	h	I	C	TFP	Y/h	Wh/Y	π	R
u	73.71	58.51	47.72	62.09	20.38	5.86	23.91	27.02	6.96	22.27
Y	56.24	80.13	44.73	67.13	33.03	4.24	41.31	40.20	10.47	16.89
h	49.84	47.54	70.45	47.99	21.78	11.62	22.61	19.47	7.23	22.38
I	59.03	66.60	45.20	80.29	19.01	3.81	33.74	36.44	7.69	21.51
C	19.19	31.59	20.15	17.10	68.30	1.57	12.93	10.31	9.93	4.50

Results: summary

1. The procedure identifies a **common shock** to y, i, c, u, ℓ
   - "Interchangeability": no matter what variable is targeted, we $\approx$ recover the same shock
   - Note: this result is somewhat less pronounced for c than for the others
2. This identified common shock has some **particular features**:
   - At business-cycle frequencies, it shows little comovement with TFP. Consistent with the semi-structural TFP shock results we discussed earlier.
   - It has little effect on long-run fluctuations of y, i, c, u
   - It is largely disconnected from inflation dynamics

Interpretation

Why are those findings useful as a background for structural business-cycle analysis?

1. The data are not inconsistent with **single-shock theories** for cyclical fluctuations
   - This single shock should lead to co-movements of standard real variables, without affecting TFP or prices much
   - Lucas (1977): *"[W]ith respect to the qualitative behavior of comovements among series, business cycles are all alike."*
2. Several candidates for this main business-cycle driver are **ruled out**
   - TFP (news) shocks would map into fluctuations in TFP (which we don't see)
     Same for many typical financial or uncertainty shocks—they map into TFP "wedges".
   - Demand shocks [+ non-rigid prices] would lead to y-π comovements (which we don't see)
3. Even **multi-shock accounts** of cycles should be consistent with the anatomy as a **reduced-form moment** [Will see next: this is an informative test.]

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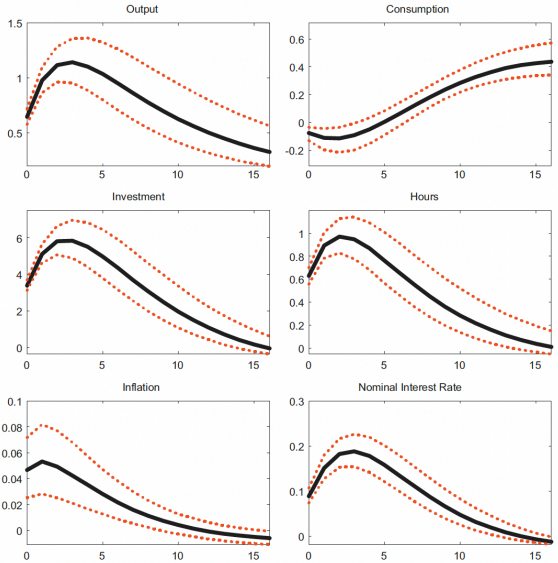
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Estimated NK models

- Let's now consider results from **structural NK model estimation**
  - Environment: RBC core + various frictions (nominal rigidities, adjustment costs, ...)
  - Then add various candidates for "shocks" as business-cycle drivers (TFP, investment technology, policy, ...) and try to account for agg. fluctuations in a likelihood sense
  - **Q**: which shocks are picked out as main cyclical drivers?
    Key logic: will pick the shocks that generate "business cycle-like" impulse responses.

- Will look at two prominent contributions to the literature:
  1. Justiniano-Primiceri-Tambalotti (2010): canonical estimated RANK model
  2. Auclert-Rognlie-Straub (2020): JPT + high household MPCs ("HANK")

Justiniano et al. (2010): investment shock IRFs



Justiniano et al. (2010): cycle decomposition

Table 1
Posterior variance decomposition at business cycle frequencies in the baseline model.

Series\shock	Medians and [5th, 95th] percentiles						
	Policy	**Neutral**	**Government**	**Investment**	**Price mark-up**	**Wage mark-up**	**Preference**
Output	0.05 [0.03, 0.08]	0.25 [0.19, 0.33]	0.02 [0.01, 0.02]	0.50 [0.42, 0.59]	0.05 [0.03, 0.07]	0.05 [0.03, 0.08]	0.07 [0.05, 0.10]
Consumption	0.02 [0.01, 0.04]	0.26 [0.20, 0.32]	0.02 [0.02, 0.03]	0.09 [0.04, 0.16]	0.01 [0.00, 0.01]	0.07 [0.04, 0.12]	0.52 [0.42, 0.61]
Investment	0.03 [0.02, 0.04]	0.06 [0.04, 0.10]	0.00 [0.00, 0.00]	0.83 [0.76, 0.89]	0.04 [0.02, 0.06]	0.01 [0.01, 0.02]	0.02 [0.01, 0.04]
Hours	0.07 [0.04, 0.10]	0.1 [0.08, 0.13]	0.02 [0.02, 0.03]	0.59 [0.52, 0.66]	0.06 [0.04, 0.09]	0.07 [0.04, 0.11]	0.08 [0.06, 0.12]
Wages	0.00 [0.00, 0.01]	0.4 [0.30, 0.52]	0.00 [0.00, 0.00]	0.04 [0.02, 0.07]	0.31 [0.23, 0.41]	0.23 [0.16, 0.32]	0.00 [0.00, 0.01]
Inflation	0.03 [0.02, 0.06]	0.14 [0.09, 0.21]	0.00 [0.00, 0.00]	0.06 [0.02, 0.13]	0.39 [0.29, 0.50]	0.34 [0.26, 0.42]	0.02 [0.01, 0.04]
Interest rates	0.17 [0.13, 0.22]	0.09 [0.06, 0.12]	0.01 [0.00, 0.01]	0.47 [0.37, 0.56]	0.05 [0.03, 0.07]	0.04 [0.03, 0.07]	0.16 [0.11, 0.23]

Note: Business cycle frequencies correspond to periodic components with cycles between 6 and 32 quarters. The decomposition is obtained using the spectrum of the DSGE model and an inverse first difference filter for output, consumption, investment and wages to reconstruct the levels. The spectral density is computed from the state space representation of the model with 500 bins for frequencies covering that range of periodicities. Medians need not add up to one.

Main result: investment technology shock as important cyclical driver

Discussion

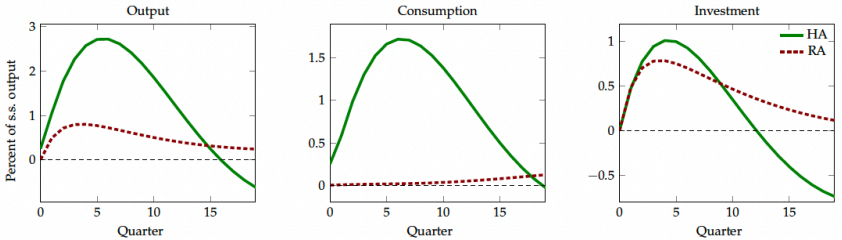
- Why does the **investment shock** play such a big role?
  - It wouldn't in vanilla RBCs: better investment opportunities $\rightarrow$ save today $\rightarrow$ $c \downarrow\downarrow$ today
    Note: sometimes referred to as "Barro-King puzzle". See RBC discussion at the beginning.
  - Here various model features help: habit formation leads to slow movements of c, and sticky prices/variable capacity utilization enable $i \uparrow$ to be largely accommodated by $y \uparrow$
- Yet it still does not look like the **main business-cycle driver** of Angeletos et al. (2020)
  - Note: consumption is essentially flat at the beginning, and the i shock accounts for little of the business-cycle fluctuations in c
  - Can formalize this through business-cycle anatomy: consumption still needs its own shock (the impatience shock), see next slide
  - Interpretation: second-moment properties of the data are matched poorly along the dimension that matters for the Angeletos et al. decomposition

Discussion


Panel A. Data, 1960–2007
Output
Consumption
Investment
Hours worked
Inflation rate
Panel B. JPT
Output
Consumption
Investment
Hours worked
Inflation rate
Y shock
I shock
h shock
C shock

Auclert et al. (2020): investment shock IRFs

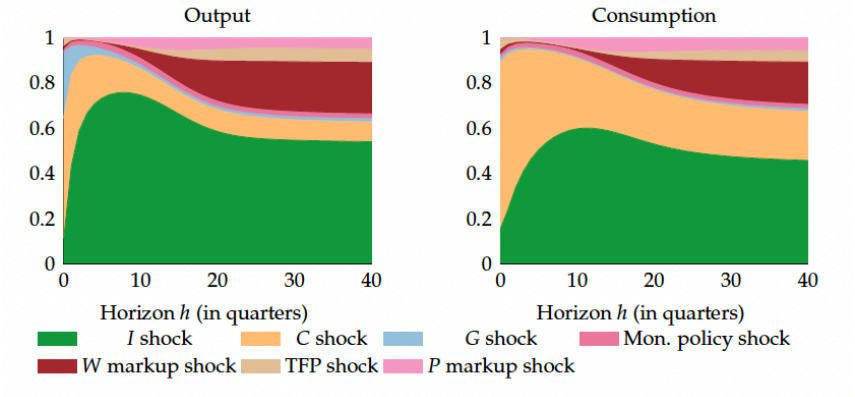
Figure 14: Impulse response to a 1-standard deviation investment shock



Note. Impulse responses are to a one-time negative shock to risk premia ϵ_t^I, using the estimated standard deviation and persistence.

Mechanism: high MPCs endogenously tie consumption to (labor) income

Auclert et al. (2020): cycle decomposition



Main result: due to high MPCs, IST shock can become the key driver of the cycle

Conjecture: their model probably also passes the Angeletos et al. "interchangeability" test.

The role of mark-up shocks

- Secondary finding: importance of **wage & price mark-up shocks**
  - In most estimated DSGE models wage/price mark-up shocks are needed to match inflation dynamics, and also matter for output at medium frequencies
  - Not surprising: labor wedge matters in "business-cycle accounting" [more on next slide] using prototype neoclassical economy, so it also matters in NK-DSGEs
- Concern for policy analysis: these "labor wedge" shocks are not likely to be structural & admit many different interpretations [preference, union bargaining, ...]
  - This is the main point of Chari-Kehoe-McGrattan (2009), and why they conclude that NK models are not yet useful for policy analysis
  - Similarly concerning for "business-cycle origin" exercises: objective is to decompose the cycle into primitive, structurally interpretable shocks, not wedges
    Of course we saw last time how this is no problem for certain counterfactual exercises. For those just need forecasts, no need to know *why* inflation moves. ...

Wedge accounting

- A different approach is proposed by Chari-Kehoe-McGrattan (2007): **wedge accounting**
  - Basic idea: fix a "prototype model" [neoclassical growth model] + historical time series that you want to explain
  - Then: back out sequence of wedges to model equations to match the target series
    Wedges: efficiency/TFP, labor, investment, and government consumption.
  - Interpretation: teaches you where to deviate from the prototype model – either in terms of shocks or by enriching endogenous propagation – to match the data
- How does this relate to the **structural approaches** above?
  - Wedge interpretation: (HA)NK-DSGE is now the prototype model, what kind of wedges do we need to account for business-cycle fluctuations?
  - With this interpretation we can think of models of business-cycle fluctuations as just another (reduced-form) slicing of data, now just relative to a particular model

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1. Semi-Structural Methods
   - TFP Shocks
   - Investment Technology Shocks

2. Structural Model Estimation
   - Basics of Likelihood-Based Estimation
   - Business-Cycle Anatomy
   - RANK & HANK Model Estimation

3. Summary

Summary

- **Main takeaways**
  1. Data are not inconsistent with a single (type of) shock as the main cyclical driver
  2. Classical TFP shocks don't work, but investment technology/demand-type shocks do
  3. Promise of HA: tie consumption endogenously to the cycle
- Note: hard to credibly establish much here using our purely semi-structural approaches. Tend to still need quite a bit of **model structure** for business-cycle origin exercises.
  - But then: do we believe that the entirety of the model & shocks is well-specified?
  - Less ambitious: wedge interpretation to guide us on model development
    E.g., build a prototype model that is consistent with empirical evidence on policy transmission. Then, relative to that model, as what kind of wedges do business cycles appear?